

Empirical Analysis of AI Models in STEM Education Using Machine Learning Techniques and Survey Insights

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Abstract—The research paper mentions the mixed-method used in the empirical evaluation of the AI tools, such as an AI-enabled tutor approach, the Platform used for adaptive learning, Virtual Labs, and an NLP assessment system. All these tools are integrated in the STEM classrooms. The blend of this study was seen during the survey and during Machine learning classification. Using 500 responses, we analyzed the data using three models, namely, Decision Tree, Logistic Regression and Random Forest. Out of the three, Random Forest achieved, highest accuracy of 89.2%, where it identified the feedback quality, relevance of the content and learner satisfaction. To achieve higher accuracy, a hybrid stack approach with a better accuracy percentage of 91.3%. In the paper, a structured framework is proposed which shows how AI-integrated classrooms emphasize in appropriate tool selection at the given time, and also explains ethical integration.

Keywords—Artificial Intelligence, Hybrid Stack, Logistic Regression, Random Forest, STEM education, NLP assessment.

I. INTRODUCTION

AI integration with STEM education has brought significant changes in the education sector. The main important change it has brought is to overcome the limitations of the traditional teaching methods. An AI-powered personalized learning method provides adaptive content delivery and real-time feedback. Most of the advanced tools, such as AI-enabled tutor systems, Virtual Labs, and NLP-oriented assessment systems, enhance the learning engagement of the students and improve their performance [1] - [2].

Even after various advancements made towards AI in STEM education, there exists still scope for some improvement. However, there are very few studies based on multiple AI-based evaluation models using the perception-based metrics [3] - [5]. Additionally, these models evaluate student perception and influence the key educational outcomes.

To reduce the gap, the current study proposes four AI-based models that are integrated. This combined method provides the student perception survey carried out with ML-based classification techniques, including Decision Trees, Random Forest, and Logistic Regression. It not only compares model performance but also provides feature importance insights and proposes a practical framework for adopting AI into STEM classrooms.

Unlike previous studies focusing on perception analysis or predictive modeling in isolation, the current work uniquely combines mixed-method empirical evaluation and SHAP-based explainability to evaluate Artificial Intelligence-enabled learning tools in STEM education. While standard classifiers are being used, the integration of explainable AI within a perception-driven evaluation framework offers a novel contribution that bridges the research and educational practice.

The remainder of the paper is organized as follows: Section II reviews various related works on AI integration into STEM education; Section III outlines the research methodology; Section IV presents the results and discusses them; Section V introduces the proposed framework of integrating Artificial Intelligence into STEM education; and finally, Section VI concludes with findings and future directions.

II. RELATED WORKS

Several studies have explored integrating AI tools into STEM education. Almutairi et al. [1] have proposed an adaptive learning system, where reinforcement learning is used to personalize the content for every learner, thus improving students' performance. According to León et al. [2], empirical research on different Artificial Intelligence models integrated into STEM learning has been conducted, thus focusing on explainable and user-centric systems.

Different ML techniques have been used to evaluate the various learning outcomes. According to Kumar et al. [3], a hybrid CNN-LSTM model was used to classify student

engagement patterns from online learning logs. Also, from another study carried out by Wang and Zhang [4]. They have applied the Random Forests algorithm to find the learning retention, mentioning the feature selection as pivotal for interpretability.

Other studies include finding the students' perceptions on AI-based learning tools. Chen et al. [5] have analysed over a thousand responses to find the satisfaction with virtual labs, identifying usability and content clarity as the top problems. Mehta and Ramesh [6] have fully emphasized the role of feedback systems with the help of NLP for motivating students. Explainable AI (XAI) has also gained prominence in education. Shukla and Banerjee [7] have implemented interpretable decision trees for academic counselling systems, while Rai et al. [8] have introduced a dashboard where XAI models are integrated to aid faculty in evaluating student performance trends.

Multiple studies have mentioned the effectiveness of ITS and the adaptive learning framework. Fernandez et al. [9] evaluated ITS with Bayesian networks, noting accuracy but poor transparency. Li et al. [10] proposed a hybrid rule-based ITS with improved interpretability. From a pedagogical point of view, Sharma and Gupta [11] in their article explained how AI tools are used to change the traditional assessment approach. Awasthi et al. [12] has designed AI-enabled teaching environment to teach subjects like engineering design that achieves positive outcomes from the students. Recent survey carried out by Dinesh et al. [13] and Fariha et al. [14] also mentioned about supervised learning approach used for understand the sentiment and continuous engagement from the students using MOOCs platform. The research findings explain the model's relevance and domain's relevance. However, many research works also suffer from other limitations such as dependence on the isolated scenario without using cross tool and secondly the lack of combined approach between perception and performance and thirdly use of the black box model that hides the pedagogical transparency.

Thus, this research addresses the gap in the mixed-method approach used in the study, which involves perception and AI validation using the four types of AI-oriented learning techniques.

III. METHODOLOGY

The study uses the mixed-method technique that is combining the data collected through surveys, using supervised ML and employing explainable AI to find out the student's satisfaction using the AI enabled tools. The same is illustrated in the Fig. 1 below.

A. Survey Design and Data Collection

A structured questionnaire is designed to evaluate the perception of the different AI based tools such as AI enabled tutor approach, AI adapted learning tools, Virtual labs and NLP oriented evaluation techniques. The survey tells that the student's using this system will have good feedback like engagement, proper relevant content, appropriate tool usage and holistic satisfaction [15], [16]. There are total of 500 responses being collected from students and faculties using

STEM disciplines, also ensuring that the sample responses are balanced by using stratified sampling approach.

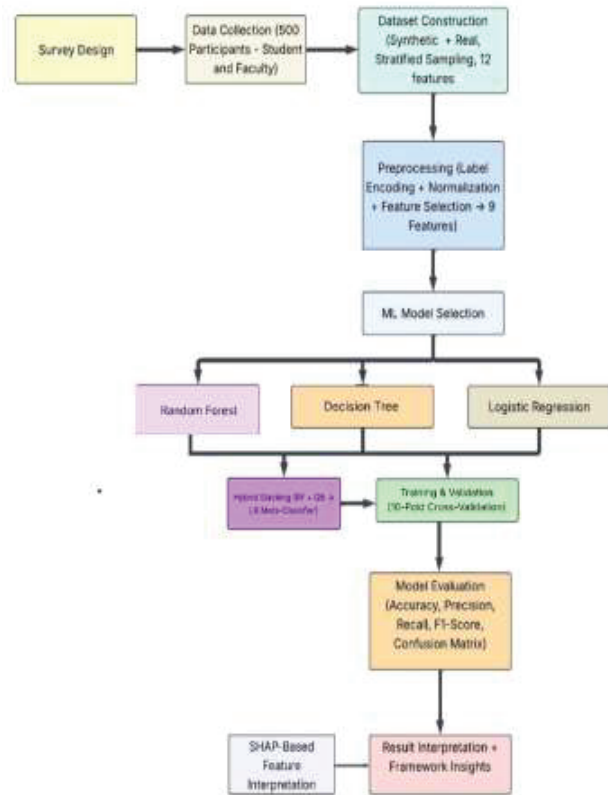


Fig. 1. Workflow for Comparative Evaluation of AI Models in STEM Education

B. Dataset Construction and Preprocessing

Response data collected are transformed into structured dataset containing 12 features. The categorical variables are labelled and numbered using normalized z-score scaling method. The synthetic data is also generated in addition to this [17], [18]. Gaussian noise based augmentation is done to remove the noise in the data where the mean value is 0 with the standard deviation of 0.5. For the categorical variables, the bootstrapping technique is being used to show the distribution of the data within the same class or category. Along with this the synthetic data is also configured with KNN value of 5 and with sampling strategy of 1.0. This reduces the noise in the data. The approach also ensures the duplication, that means there are some data with similar or same results across samples [19], [20]. Feature selections were being used via univariate statistical method, where the dataset is reduced to 9 key predictors using 80 : 20 training to testing data ratio split.

C. Machine Learning Models

We have used three supervised ML classifiers to analyze the student's satisfaction:

- Logistic Regression (LR) as a linear baseline,
- Decision Tree (DT) for rule-based interpretability,

- Random Forest (RF) for robust ensemble learning.

The above mentioned ML models were used for the predicting perception of the students. The traditional ML models like Decision Tree, Log. Regression and the Random Forest are used for the robustness. Here the novelty lies in the use of mixed method AI driven technique used. The embedding of SHAP into system provides the transparency which is rarely found in the ML based educational research.

D. Hybrid Ensemble Framework

An ensemble hybrid approach is used to boost the predictive performance and also bringing the novelty in approach by integrating all the three ML models Random Forst, Gradient Boosting and Logistic Regression.

E. Model Training and Validation

Models that are implemented using scikit learn library of the Python language are trained by using 10-fold cross validation to ensure normalized data. Each fold of the data preserves the class balance and ultimately the final metrics is calculated to find the average of all the 10 folds. Among the data, 20% of it was reserved for the sample validation.

F. Explainable AI with SHAP

For feature contributions, SHAP or Shapley Additive exPlanations are being used to train the Random Forest model. The SHAP reveals the quality of the feedback from the students, relevance of the contents and proper use of the tool, which is used for top prediction parameters for the satisfaction rate. The ML insights supports the learner-centric AI integrated approach.

G. System Requirements

All the experiments that are conducted on the system with the required hardware configurations of Intel i7 core, with the memory of 16 GB and Windows 11 of 64-bit OS. The Python version 3.11 or above with sci-kit learn library. Along with other libraries like SHAP, Pandas, NumPy and graphical libraries like Matplotlib. For our research no GPUs were used.

H. Evaluation Metrics

The performance of the three ML models, such as LR, DT, and RF, was evaluated based on five critical metrics: accuracy of the Model, precision of the Model, recall, F1-score, and confusion matrix. Out of the three models, the RF model provided the best performance, achieving an accuracy of 89.2%, followed by the DT Model (83.6%) and last with the LR (81.4%). A fourth ensemble model, a “Hybrid stacking ensemble” combining the three ML models as a meta-classifier, was also evaluated. Table I below shows how the proposed hybrid model outshines the other classifiers by achieving a maximum accuracy of 91.3%. The result also showed an improved F1 score and reduced misclassification rates. This shows the benefit of combining RF and DT with an LR to capture complex and generalizable patterns in learner perception data. To validate this, the model was compared against the recent studies carried out by Kumar et.al. [3], where the CNN-LSTM model showed 88% accuracy, and another study by Zhang [4] reported that the Random Forest Model showed 87% accuracy. The proposed

hybrid ensemble model has outperformed both with 91.3% accuracy and an F1-score of 0.88, suggesting better generalization and prediction consistency. A summary of different model performances is shown in Table I.

TABLE I. PERFORMANCE COMPARISON OF MACHINE LEARNING CLASSIFIERS ON LEARNER SATISFACTION PREDICTION.

Model	Accuracy (%)	Precision	Recall	F1-Score
Logistic Regression	81.4	0.78	0.76	0.77
Decision Tree	83.6	0.81	0.8	0.8
Random Forest	89.2	0.87	0.86	0.86
Hybrid Ensemble	91.3	0.89	0.88	0.88

Table I shows Hybrid Ensemble’s dominance over the other ML models by achieving the highest accuracy and F1-score. The confusion matrix in Fig. 2 shows that Hybrid Ensemble yielded the lowest number of misclassifications, with high true positive and true negative rates. This validates the model's capability to generalize well to unseen data. Logistic Regression showed limitations in capturing non-linear patterns, while Decision Tree performed moderately but was slightly prone to overfitting. Random Forest, on the other hand, had better results, but the Hybrid ensemble approach has improved the general limitations over the other three models.

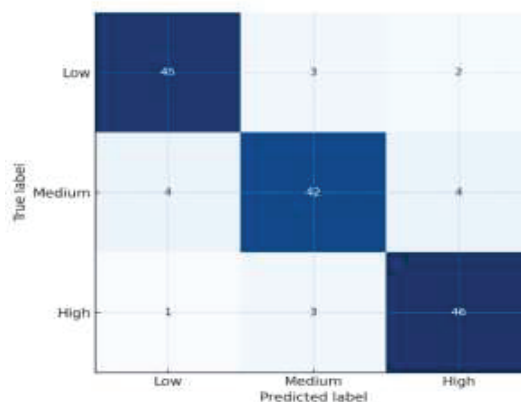


Fig. 2. Confusion matrix for the Ensembled model – Hybrid approach

I. Explainability with SHAP Values

To improve the transparency of the model and support the insight SHAP analysis has been conducted on the ensemble hybrid ML model. SHAP is used to know the features of the model’s predictions at different levels. The summary of SHAP data gives the feedback quality, relevance of the content towards the domain and usage of tools helps in predicting the student’s learning satisfaction. This explainability layer evaluates the model’s behavior and strengthens the trustworthiness in the framework. The integration of SHAP into AI integrated model supports the educational institutions for decision making. As per the Fig. 3, the quality of the feedback, relevance of the content

towards the domain and usage of the tools as the top insight contributors.

J. Framework Insights

Based on the various findings of the model's performances and SHAP interpretation, a data driven decision making is proposed. This proposed framework ensures the prioritization in the feedback, improvement in the design content, improving AI based education tools. The insights also show the refinement tools like AI based tutor system, Virtual labs that ensures the students engagement is in-aligned with continuous learning. The system provides the strategic method of institutions using AI tools in the STEM education.

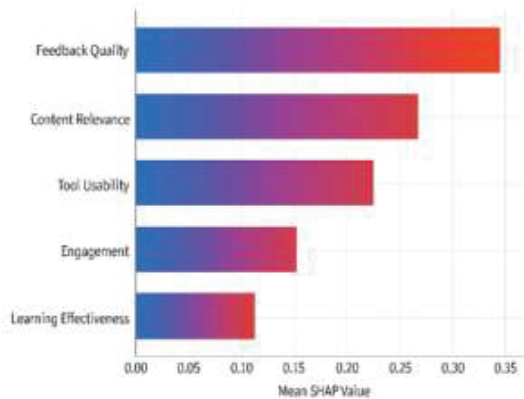


Fig. 3. SHAP summary plot values

The comparative insights of all the AI tools mentioned in the paper reveals the usage patterns of the students. Among all the approach, the virtual labs have received better feedback whereas NLP based evaluation tool have shown relatively lesser satisfaction levels. AI based tutor system has received response for good feedback integration. Whereas the Adaptive based learning software has shown promise towards personalized content providing to the users. The insight tells what improvements must be used in deployment and in the training [21]-[22].

IV. PROPOSED FRAMEWORK FOR AI INTEGRATION IN STEM EDUCATION

Based on the various studies, a practical and ethical approach has been proposed in the study that shows how AI-enabled learning models are used in the institutions. The proposed approach combines all the aspects like model performance, perception from the students that is required in the deployment [23].

A. Model Selection Criteria

- Perception-Evaluated Effectiveness: There is a need to give AI tools a higher satisfaction rate in the student's feedback approach especially in proving

the proper content relevance, ease of use of the content and adaptiveness.

- Interpretable Performance: Decision Trees or other XAI-compatible models should be preferred for their transparency in prediction, enabling meaningful academic insights.
- Domain Suitability: Models should be matched to domain complexity (e.g., virtual labs in engineering, ITS for problem-solving disciplines) [24].

B. Deployment Strategy

- Pilot Implementation: Conduct controlled pilots using selected AI tools in diverse classrooms. Collect real-time perception and performance data.
- Feedback Loop and Monitoring: Analyze post-implementation feedback through lightweight ML models to track perception shifts and effectiveness.
- Faculty and Admin Involvement: Different academic staff should participate in tool evaluation and configuration, ensuring pedagogical alignment.

C. Ethical and Institutional Oversight

- Ensure compliance with data privacy policies and anonymized data processing.
- Use explainable models to ensure trust among faculty and students.
- Periodic audits should be conducted to monitor bias, fairness, and inclusivity.

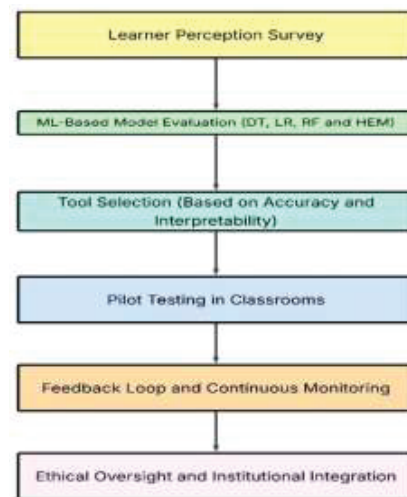


Fig. 4. Framework for Responsible AI Model Adoption in STEM Classrooms

D. Visual Workflow of the Framework

In the Fig.4 the AI adapted model in the STEM classrooms is proposed. The flow diagram shows the workflow of how AI is used in the STEM education.

V. CONCLUSION AND FUTURE WORK

The study in the paper proposes the mixed-method approach of integrating different ML models to evaluate the student's satisfaction level in using the AI-based tool for the STEM education. The survey data shows among the different ML models Hybrid Ensemble model has achieved best accuracy of 91.3%. SHAP analysis is used in identifying the quality of the feedback, relevance of the content and right tool usage. The AI based framework provides the insight for every stakeholder of the educational institution in evaluating AI tools strengths and weakness. The major limitation of the study is that the data used is sourced against the single institution, to have better accuracy it is required to use the data with different demographics, which gives the analysis of usage of AI tools across different institutions, regions, type of courses. All this put together will give the perception scores, with each of it carrying the weightage. Alternatively, the study does not generalize the pedagogical practices across demographics. The future studies can be used to combine all the data from different institutions and validate the same to strengthen proposed approach.

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